

10.1 Clustering is Grouping

If we group classification procedures based on whether they can generate prules after their training, then SpecToGen, GenToSpec, decision tree, and evolutionary procedures (Chapters 1 to 4) fall into the group of generating prules. Bayes, nearest neighbour, and neural nets (Chapters 5 to 7) fall into the group of not generating prules. The generation of prules serves as an attribute based on which we have carried out the grouping: members of a group are similar for that attribute. If, however, we group the classification procedures based on whether the procedure can be said to model the brain or whether they can be said to model the mind, then neural nets fall into the group modelling the brain, and the remaining procedures fall into the group modelling the mind. Grouping can be carried out based on more than one attribute. If we group the classification procedures on whether they generate prules after training as well as whether they model the brain or the mind, then we get three groups: (a) SpecToGen, GenToSpec, decision tree, and evolutionary procedures which model the mind and generate prules; (b) Bayes and nearest neighbour procedures which model the mind and do not generate prules; and (c) neural net which models the brain but does not generate prules.

Patterns can also be put into groups based on the values of their attributes. Such groups are called *clusters*, and the process of forming clusters is called *clustering*. The process has often been found to be useful in the exploratory stages of researching a domain to learn how patterns in the domain can be clustered. Clustering is also known as 'unsupervised learning', that is, there

NAME of pattern	Values of attributes	
	A_1	A_2
X1	1	1
X2	2	3
X3	3	1
X4	4	4
X5	5	2

FIGURE 10.1 Set of five patterns that are to be clustered.

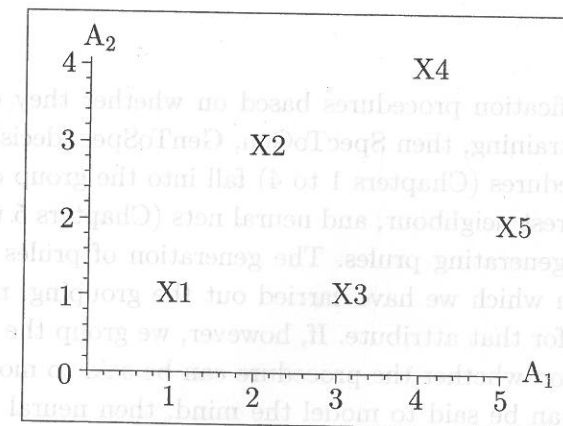


FIGURE 10.2 Representing each pattern given in Fig. 10.1 as a point in the A_1 - A_2 coordinate space.

is no training set available to supervise, or guide, the learning. After we have clustered some given patterns, we may consider each cluster to be a class.

Study the five patterns X1 to X5 and the values of their numeric attributes A_1 and A_2 in Fig. 10.1. It is not necessary to know what the attributes mean; only their values need to be considered. Intuitively put the five patterns into two clusters such that the patterns within a cluster are more similar to one another than to patterns in another cluster. It might help to first represent each pattern as a point in the A_1 - A_2 coordinate space as shown in Fig. 10.2; points in a cluster are closer to one another than to points in another cluster. Once you have finished clustering, ask a friend to form the clusters. It is possible that the clusters made by your friend are different from the clusters made by you.

10.2 Agglomerative Hierarchical Clustering

There exist procedures which—given $r \geq 1$ patterns and the values of their attributes $A_1, A_2, \dots, A_M$ —can put the patterns into $k \geq 1$ clusters, where $k \leq r$. Different procedures may, however, produce different clusters. A commonly adopted procedure has the following steps.

1. Choose a value for k , which denotes the number of clusters desired.
2. Consider each of the r patterns to be in a separate cluster. In other words, each cluster contains one pattern.

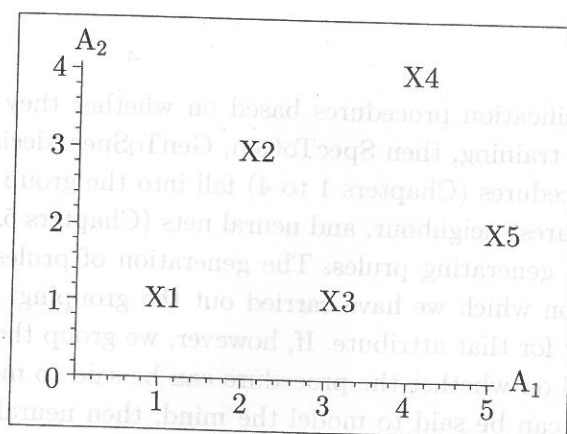


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Study the five patterns X1 to X5 and the values of their numeric attributes A_1 and A_2 in Fig. 10.1. It is not necessary to know what the attributes mean; only their values need to be considered. Intuitively put the five patterns into two clusters such that the patterns within a cluster are more similar to one another than to patterns in another cluster. It might help to first represent each pattern as a point in the A_1 - A_2 coordinate space as shown in Fig. 10.2; points in a cluster are closer to one another than to points in another cluster. Once you have finished clustering, ask a friend to form the clusters. It is possible that the clusters made by your friend are different from the clusters made by you.

10.2 Agglomerative Hierarchical Clustering

There exist procedures which—given $r \geq 1$ patterns and the values of their attributes $A_1, A_2, \dots, A_M$ —can put the patterns into $k \geq 1$ clusters, where $k \leq r$. Different procedures may, however, produce different clusters. A commonly adopted procedure has the following steps.

1. Choose a value for k , which denotes the number of clusters desired.
2. Consider each of the r patterns to be in a separate cluster. In other words, each cluster contains one pattern.

Agglomerative Hierarchical Clustering

3. If there are k clusters, then return from the procedure with the k clusters.
4. Find the pairwise distances between the clusters. Merge the two closest clusters into one cluster, thus reducing the number of clusters by one. If there is more than one pair of clusters that are the closest to each other, then arbitrarily select a pair to merge, for instance, the first such pair encountered while scanning the clusters in some order. Go to step 3.

Step 4 above measures the distance between clusters. Before deciding how to measure that, we need to first decide how to measure the distance between patterns. Euclidean, city-block, Hamming distance (explained in Sections 6.2 and 6.3), or some function of these can be chosen to measure the distance. In practice, the Euclidean distance is the one most often chosen. Figure 10.3 shows the Euclidean distances between the patterns given in Fig. 10.1, which can be confirmed from Fig. 10.2.

The distances between two clusters C' and C'' can then be measured as follows. First calculate the pairwise distance for every pattern in cluster C' with every pattern in cluster C'' . Then, the distance between cluster C' and cluster C'' can be defined to be one of the following:

- The lowest of the pairwise distances—if this definition is chosen for clustering, then the procedure is known as *single-linkage*, or *nearest neighbour*, clustering.

Patterns	X1	X2	X3	X4	X5
X1	0	2.2	2	4.2	4.1
X2	2.2	0	2.2	2.2	3.2
X3	2	2.2	0	3.2	2.2
X4	4.2	2.2	3.2	0	2.2
X5	4.1	3.2	2.2	2.2	0

FIGURE 10.3 Euclidean distances between the five patterns given in Fig. 10.1. The distance between patterns X1 and X2 is $\sqrt{(2-1)^2 + (3-1)^2} = 2.2$ (rounded off). The diagonal values are zero because the distance between a pattern and itself is zero. The table is symmetric because, for example, the distance between patterns X1 and X2 is equal to the distance between X2 and X1.

To create two clusters of the five patterns X_1 to X_5 given in Fig. 10.1 by the single-linkage procedure using Euclidean distance, we begin by creating 5 clusters $\{X_1\}$, $\{X_2\}$, $\{X_3\}$, $\{X_4\}$, and $\{X_5\}$. Considering the off-diagonal values in Fig. 10.3, we see that the minimum distance between any two clusters is 2: it is between $\{X_1\}$ and $\{X_3\}$, which we merge into one cluster $\{X_1, X_3\}$. We then obtain the distances between the four clusters $\{X_1, X_3\}$, $\{X_2\}$, $\{X_4\}$, and $\{X_5\}$, as given in the following symmetric table:

Clusters	$\{X_1, X_3\}$	$\{X_2\}$	$\{X_4\}$	$\{X_5\}$
$\{X_1, X_3\}$	0	2.2	3.2	2.2
$\{X_2\}$	2.2	0	2.2	3.2
$\{X_4\}$	3.2	2.2	0	2.2
$\{X_5\}$	2.2	3.2	2.2	0

Since we are using the single-linkage procedure, the distance between $\{X_1, X_3\}$ and $\{X_4\}$ is the lesser of the distance between X_1 and X_4 , and that between X_3 and X_4 , the two distances obtained from Fig. 10.3. Therefore, the distance between $\{X_1, X_3\}$ and $\{X_4\}$ is $\min(4.2, 3.2) = 3.2$, which is shown in the table above. The other values in the table are obtained similarly. The smallest off-diagonal value in the table is 2.2. We can merge any pair of clusters having a distance of 2.2 between them. Let us select the pair corresponding to 2.2 we encounter first when we scan the table row by row. So we merge $\{X_2\}$ with $\{X_1, X_3\}$. We now have the three clusters $\{X_1, X_2, X_3\}$, $\{X_4\}$, and $\{X_5\}$, the distances between which are in the following symmetric table:

Clusters	$\{X_1, X_2, X_3\}$	$\{X_4\}$	$\{X_5\}$
$\{X_1, X_2, X_3\}$	0	2.2	2.2
$\{X_4\}$	2.2	0	2.2
$\{X_5\}$	2.2	2.2	0

With our using the single-linkage procedure, the distance between $\{X_1, X_2, X_3\}$ and $\{X_4\}$ is the least of the distances between X_4 and either of X_1, X_2 , and X_3 (see Fig. 10.3). The distance between $\{X_1, X_2, X_3\}$ and $\{X_4\}$ accordingly is $\min(4.2, 2.2, 3.2) = 2.2$, as shown in the above table. The smallest off-diagonal value in the table is 2.2. Selecting the pair of clusters corresponding to 2.2 we encounter first when we scan the table row by row, we merge $\{X_1, X_2, X_3\}$ with $\{X_4\}$ to finally obtain these two clusters: $\{X_1, X_2, X_3, X_4\}$ and $\{X_5\}$.

FIGURE 10.4

An example of clustering the patterns X_1 to X_5 given in Fig. 10.1 by the single-linkage procedure using Euclidean distance. In the tables, the diagonal elements are zero because the distance between a cluster and itself is zero. The above tables can be compared with those given in Figs 10.5 to 10.7.

To create two clusters of the five patterns X_1 to X_5 given in Fig. 10.1 by the complete-linkage procedure using Euclidean distance, we begin by creating 5 clusters $\{X_1\}$, $\{X_2\}$, $\{X_3\}$, $\{X_4\}$, and $\{X_5\}$. Considering the off-diagonal values given in Fig. 10.3, we see that the minimum distance between any two clusters is 2: it is between $\{X_1\}$ and $\{X_3\}$, which we merge into one cluster $\{X_1, X_3\}$. We then obtain the distances between the four clusters $\{X_1, X_3\}$, $\{X_2\}$, $\{X_4\}$, and $\{X_5\}$, as given in the following symmetric table:

Clusters	$\{X_1, X_3\}$	$\{X_2\}$	$\{X_4\}$	$\{X_5\}$
$\{X_1, X_3\}$	0	2.2	4.2	4.1
$\{X_2\}$	2.2	0	2.2	3.2
$\{X_4\}$	4.2	2.2	0	2.2
$\{X_5\}$	4.1	3.2	2.2	0

Since we are using the complete-linkage procedure, the distance between $\{X_1, X_3\}$ and $\{X_4\}$ is the greater of the distance between X_1 and X_4 , and that between X_3 and X_4 , the two distances obtained from Fig. 10.3. Therefore, the distance between $\{X_1, X_3\}$ and $\{X_4\}$ is $\max(4.2, 3.2) = 4.2$, which is what is shown in the table above. The other values in the table are obtained similarly. The smallest off-diagonal value in the table is 2.2. We can merge any pair of clusters having a distance of 2.2 between them. Let us select the pair corresponding to 2.2 we encounter first when we scan the table row by row. So we merge $\{X_2\}$ with $\{X_1, X_3\}$. We now have the three clusters $\{X_1, X_2, X_3\}$, $\{X_4\}$, and $\{X_5\}$, the distances between which are in the following symmetric table:

Clusters	$\{X_1, X_2, X_3\}$	$\{X_4\}$	$\{X_5\}$
$\{X_1, X_2, X_3\}$	0	4.2	4.1
$\{X_4\}$	4.2	0	2.2
$\{X_5\}$	4.1	2.2	0

With our using the complete-linkage procedure, the distance between $\{X_1, X_2, X_3\}$ and $\{X_4\}$ is the greatest of the distances between X_4 and either of X_1, X_2 , and X_3 (see Fig. 10.3). The distance between $\{X_1, X_2, X_3\}$ and $\{X_4\}$ accordingly is $\max(4.2, 2.2, 3.2) = 4.2$, as shown in the above table. The smallest off-diagonal value in the table is 2.2. We merge $\{X_5\}$ with $\{X_4\}$ to finally obtain these two clusters: $\{X_1, X_2, X_3\}$ and $\{X_4, X_5\}$.

FIGURE 10.5

An example of clustering the patterns X_1 to X_5 given in Fig. 10.1 by the complete-linkage procedure using Euclidean distance. In the tables, the diagonal elements are zero because the distance between a cluster and itself is zero. The above tables can be compared with those given in Figs 10.4, 10.6, and 10.7.

To create two clusters of the five patterns X1 to X5 given in Fig. 10.1 by the **single-linkage** procedure using Euclidean distance, we begin by creating 5 clusters {X1}, {X2}, {X3}, {X4}, and {X5}. Considering the off-diagonal values in Fig. 10.3, we see that the minimum distance between any two clusters is 2: it is between {X1} and {X3}, which we merge into one cluster {X1, X3}. We then obtain the distances between the four clusters {X1, X3}, {X2}, {X4}, and {X5}, as given in the following symmetric table:

Clusters	{X1, X3}	{X2}	{X4}	{X5}
{X1, X3}	0	2.2	3.2	2.2
{X2}	2.2	0	2.2	3.2
{X4}	3.2	2.2	0	2.2
{X5}	2.2	3.2	2.2	0

Since we are using the single-linkage procedure, the distance between {X1, X3} and, say, {X4} is the lesser of the distance between X1 and X4, and that between X3 and X4, the two distances obtained from Fig. 10.3. Therefore, the distance between {X1, X3} and {X4} is $\min(4.2, 3.2) = 3.2$, which is what is shown in the table above. The other values in the table are obtained similarly. The smallest off-diagonal value in the table is 2.2. We can merge any pair of clusters having a distance of 2.2 between them. Let us select the pair corresponding to 2.2 we encounter first when we scan the table row by row. So we merge {X2} with {X1, X3}. We now have the three clusters {X1, X2, X3}, {X4}, and {X5}, the distances between which are in the following symmetric table:

Clusters	{X1, X2, X3}	{X4}	{X5}
{X1, X2, X3}	0	2.2	2.2
{X4}	2.2	0	2.2
{X5}	2.2	2.2	0

With our using the single-linkage procedure, the distance between {X1, X2, X3} and, say, {X4} is the least of the distances between X4 and either of X1, X2, and X3 (see Fig. 10.3). The distance between {X1, X2, X3} and {X4} accordingly is $\min(4.2, 2.2, 3.2) = 2.2$, as shown in the above table. The smallest off-diagonal value in the table is 2.2. Selecting the pair of clusters corresponding to 2.2 we encounter first when we scan the table row by row, we merge {X1, X2, X3} with {X4} to finally obtain these two clusters: {X1, X2, X3, X4} and {X5}.

FIGURE 10.4 An example of clustering the patterns X1 to X5 given in Fig. 10.1 by the single-linkage procedure using Euclidean distance. In the tables, the diagonal elements are zero because the distance between a cluster and itself is zero. The above tables can be compared with those given in Figs 10.5 to 10.7.

To create two clusters of the five patterns X1 to X5 given in Fig. 10.1 by the **complete-linkage** procedure using Euclidean distance, we begin by creating 5 clusters {X1}, {X2}, {X3}, {X4}, and {X5}. Considering the off-diagonal values given in Fig. 10.3, we see that the minimum distance between any two clusters is 2: it is between {X1} and {X3}, which we merge into one cluster {X1, X3}. We then obtain the distances between the four clusters {X1, X3}, {X2}, {X4}, and {X5}, as given in the following symmetric table:

Clusters	{X1,X3}	{X2}	{X4}	{X5}
{X1,X3}	0	2.2	4.2	4.1
{X2}	2.2	0	2.2	3.2
{X4}	4.2	2.2	0	2.2
{X5}	4.1	3.2	2.2	0

Since we are using the complete-linkage procedure, the distance between {X1, X3} and, say, {X4} is the greater of the distance between X1 and X4, and that between X3 and X4, the two distances obtained from Fig. 10.3. Therefore, the distance between {X1, X3} and {X4} is $\max(4.2, 3.2) = 4.2$, which is what is shown in the table above. The other values in the table are obtained similarly. The smallest off-diagonal value in the table is 2.2. We can merge any pair of clusters having a distance of 2.2 between them. Let us select the pair corresponding to 2.2 we encounter first when we scan the table row by row. So we merge {X2} with {X1, X3}. We now have the three clusters {X1, X2, X3}, {X4}, and {X5}, the distances between which are in the following symmetric table:

Clusters	{X1, X2, X3}	{X4}	{X5}
{X1, X2, X3}	0	4.2	4.1
{X4}	4.2	0	2.2
{X5}	4.1	2.2	0

With our using the complete-linkage procedure, the distance between {X1, X2, X3} and, say, {X4} is the greatest of the distances between X4 and either of X1, X2, and X3 (see Fig. 10.3). The distance between {X1, X2, X3} and {X4} accordingly is $\max(4.2, 2.2, 3.2) = 4.2$, as shown in the above table. The smallest off-diagonal value in the table is 2.2. We merge {X5} with {X4} to finally obtain these two clusters: {X1, X2, X3} and {X4, X5}.

FIGURE 10.5 An example of clustering the patterns X1 to X5 given in Fig. 10.1 by the complete-linkage procedure using Euclidean distance. In the tables, the diagonal elements are zero because the distance between a cluster and itself is zero. The above tables can be compared with those given in Figs 10.4, 10.6, and 10.7.

To create two clusters of the five patterns X1 to X5 given in Fig. 10.1 by the average-linkage procedure using Euclidean distance, we begin by creating 5 clusters {X1}, {X2}, {X3}, {X4}, and {X5}. Considering the off-diagonal values given in Fig. 10.3, we see that the minimum distance between any two clusters is 2: it is between {X1} and {X3}, which we merge into one cluster {X1, X3}. We then obtain the distances between the four clusters {X1, X3}, {X2}, {X4}, and {X5}, as given in the following symmetric table:

Clusters	{X1, X3}	{X2}	{X4}	{X5}
{X1, X3}	0	2.2	3.7	3.2
{X2}	2.2	0	2.2	2.2
{X4}	3.7	2.2	0	2.2
{X5}	3.2	2.2	2.2	0

Since we are using the average-linkage procedure, the distance between {X1, X3} and {X4}, the last two distances obtained from Fig. 10.3. Therefore, the distance between X4, the last two distances obtained from Fig. 10.3, which is what is shown in the table above. The other values in the table are obtained similarly. For simplicity, the values have been rounded off to one place after the decimal. The smallest off-diagonal value in the table is 2.2. We can merge any pair of clusters having a distance of 2.2 between them. Let us select the pair corresponding to 2.2 we encounter first when we scan the table row by row. So we merge {X2} with {X1, X3}. We now have the three clusters {X1, X2, X3}, {X4}, and {X5}, the distances between which are in the following symmetric table:

Clusters	{X1, X2, X3}	{X4}	{X5}
{X1, X2, X3}	0	3.2	3.2
{X4}	3.2	0	2.2
{X5}	3.2	2.2	0

With our using the average-linkage procedure, the distance between {X1, X2, X3} and, say, {X4} is the average of the distances between X4 and each of X1, X2, and X3 (see Fig. 10.3). The distance between {X1, X2, X3} and {X4} accordingly is $(4.2 + 3.2 + 2.2)/3 = 3.2$, as shown in the above table. The smallest off-diagonal value in the table is 2.2. We merge {X5} with {X4} to finally obtain these two clusters: {X1, X2, X3} and {X4, X5}.

FIGURE 10.6

An example of clustering the patterns X1 to X5 given in Fig. 10.1 by the average-linkage procedure using Euclidean distance. In the tables, the diagonal elements are zero because the distance between a cluster and itself is zero. The above tables can be compared with those given in Figs 10.4, 10.5, and 10.7.

- The highest of the pairwise distances—if this definition is chosen, then the procedure is known as *complete-linkage*, or *farthest neighbour*, clustering.
- The average of the pairwise distances—if this definition is chosen, then the procedure is known as *average-linkage* clustering.

In addition to this, there is one more way to measure the distance between clusters. The distance between their centroids. For $1 \leq i \leq M$, the value of the i th coordinate of the centroid of a cluster is the mean of the values of attribute A_i of the patterns in the cluster (this is what you did for a prototype in Section 6.5). If this definition is chosen to measure the distance between clusters, then the clustering procedure is known as *centroid-linkage* clustering.

As examples, the patterns given in Fig. 10.1 are clustered using Euclidean distance by single-linkage clustering in Fig. 10.4, by complete-linkage clustering in Fig. 10.5, by average-linkage clustering in Fig. 10.6, and by centroid-linkage clustering in Fig. 10.7. The four procedures do not produce identical clusters. For calculating the distance between clusters, the single-linkage and complete-linkage clustering procedures are dependent on single patterns in clusters. An outlier (defined in Section 6.1) in either of the clusters can affect the distance measured between the two clusters. Accordingly, it is recommended that the average-linkage or the centroid-linkage clustering procedures be preferred over the single-linkage and complete-linkage clustering procedures.

All the above four procedures are said to be *agglomerative hierarchical* clustering procedures. They are agglomerative because they build larger clusters from smaller clusters, and they are hierarchical because the larger clusters are built by merging the smaller clusters.

10.3 k-means Clustering

follows:

1. Consider k of the r given patterns to each form a cluster. We thus have k clusters, each with one pattern. The remaining patterns remain as

To create two clusters of the five patterns X1 to X5 given in Fig. 10.1 by the **average-linkage** procedure using Euclidean distance, we begin by creating 5 clusters {X1}, {X2}, {X3}, {X4}, and {X5}. Considering the off-diagonal values given in Fig. 10.3, we see that the minimum distance between any two clusters is 2: it is between {X1} and {X3}, which we merge into one cluster {X1, X3}. We then obtain the distances between the four clusters {X1, X3}, {X2}, {X4}, and {X5}, as given in the following symmetric table:

Clusters	{X1, X3}	{X2}	{X4}	{X5}
{X1, X3}	0	2.2	3.7	3.2
{X2}	2.2	0	2.2	3.2
{X4}	3.7	2.2	0	2.2
{X5}	3.2	3.2	2.2	0

Since we are using the average-linkage procedure, the distance between {X1, X3} and, say, {X4} is the average of the distance between X1 and X4, and that between X3 and X4, the last two distances obtained from Fig. 10.3. Therefore, the distance between {X1, X3} and {X4} is $(4.2 + 3.2)/2 = 3.7$, which is what is shown in the table above. The other values in the table are obtained similarly. For simplicity, the values have been rounded off to one place after the decimal. The smallest off-diagonal value in the table is 2.2. We can merge any pair of clusters having a distance of 2.2 between them. Let us select the pair corresponding to 2.2 we encounter first when we scan the table row by row. So we merge {X2} with {X1, X3}. We now have the three clusters {X1, X2, X3}, {X4}, and {X5}, the distances between which are in the following symmetric table:

Clusters	{X1, X2, X3}	{X4}	{X5}
{X1, X2, X3}	0	3.2	3.2
{X4}	3.2	0	2.2
{X5}	3.2	2.2	0

With our using the average-linkage procedure, the distance between {X1, X2, X3} and, say, {X4} is the average of the distances between X4 and each of X1, X2, and X3 (see Fig. 10.3). The distance between {X1, X2, X3} and {X4} accordingly is $(4.2 + 3.2 + 2.2)/3 = 3.2$, as shown in the above table. The smallest off-diagonal value in the table is 2.2. We merge {X5} with {X4} to finally obtain these two clusters: {X1, X2, X3} and {X4, X5}.

FIGURE 10.6 An example of clustering the patterns X1 to X5 given in Fig. 10.1 by the average-linkage procedure using Euclidean distance. In the tables, the diagonal elements are zero because the distance between a cluster and itself is zero. The above tables can be compared with those given in Figs 10.4, 10.5, and 10.7.

- The highest of the pairwise distances—if this definition is chosen, then the procedure is known as *complete-linkage*, or *farthest neighbour*, clustering.
- The average of the pairwise distances—if this definition is chosen, then the procedure is known as *average-linkage* clustering.

In addition to this, there is one more way to measure the distance between clusters. The distance between clusters C' and C'' is defined to be equal to the distance between their centroids. For $1 \leq i \leq M$, the value of the A_i th coordinate of the centroid of a cluster is the mean of the values of attribute A_i of the patterns in the cluster (this is what you did for a prototype in Section 6.5). If this definition is chosen to measure the distance between clusters, then the clustering procedure is known as *centroid-linkage* clustering.

As examples, the patterns given in Fig. 10.1 are clustered using Euclidean distance by single-linkage clustering in Fig. 10.4, by complete-linkage clustering in Fig. 10.5, by average-linkage clustering in Fig. 10.6, and by centroid-linkage clustering in Fig. 10.7. The four procedures do not produce identical clusters. For calculating the distance between clusters, the single-linkage and complete-linkage clustering procedures are dependent on single patterns in clusters. An outlier (defined in Section 6.1) in either of the clusters can affect the distance measured between the two clusters. Accordingly, it is recommended that the average-linkage or the centroid-linkage clustering procedures be preferred over the single-linkage and complete-linkage clustering procedures.

All the above four procedures are said to be *agglomerative hierarchical* clustering procedures. They are agglomerative because they build larger clusters from smaller clusters, and they are hierarchical because the larger clusters are built by merging the smaller clusters.

10.3 k-means Clustering

A non-hierarchical way of clustering is the *k-means* procedure, which is as follows:

1. Consider k of the r given patterns to each form a cluster. We thus have k clusters, each with one pattern. The remaining patterns remain as

To create two clusters of the five patterns X1 to X5 given in Fig. 10.1 by the centroid-linkage procedure using Euclidean distance, we begin by creating 5 clusters {X1}, {X2}, {X3}, {X4}, and {X5}. Considering the off-diagonal values given in Fig. 10.3, we see that the minimum distance between any two clusters is 2: it is between {X1} and {X3}, which we merge into one cluster. The coordinates of the centroid of the resulting cluster {X1, X3} are $[(1 + 3)/2 = 2, (1 + 1)/2 = 1]$. We then obtain the distances between the four clusters {X1, X3}, {X2}, {X4}, and {X5}, as given in the following symmetric table:

Clusters	{X1, X3}	{X2}	{X4}	{X5}
{X1, X3}	0	2	3.6	3.2
{X2}	2	0	2.2	3.2
{X4}	3.6	2.2	0	2.2
{X5}	3.2	3.2	2.2	0

Since we are using the centroid-linkage procedure, the distance between {X1, X3} and, say, {X4} is $\sqrt{(4 - 2)^2 + (4 - 1)^2} = 3.6$, which is what is shown in the table above. The other values in the table are obtained similarly. For simplicity, the values have been rounded off to one place after the decimal. The smallest off-diagonal value in the table is 2; it is between {X1, X3} and {X2}, which we merge into one cluster. The coordinates of the centroid of the resulting cluster {X1, X2, X3} are $[(1 + 2 + 3)/3 = 2, (1 + 3 + 1)/3 = 1.67]$. We now have the three clusters {X1, X2, X3}, {X4}, and {X5}, the distances between which are in the following symmetric table:

Clusters	{X1, X2, X3}	{X4}	{X5}
{X1, X2, X3}	0	3.1	3
{X4}	3.1	0	2.2
{X5}	3	2.2	0

With our using the centroid-linkage procedure, the distance between {X1, X2, X3} and, say, {X4} is $\sqrt{(4 - 2)^2 + (4 - 1.67)^2} = 3.1$, as shown in the above table. The smallest off-diagonal value in the table is 2.2. We merge {X5} with {X4} to finally obtain these two clusters: {X1, X2, X3} and {X4, X5}.

FIGURE 10.7

An example of clustering the patterns X1 to X5 given in Fig. 10.1 by the centroid-linkage procedure using Euclidean distance. In the tables, the diagonal elements are zero because the distance between a cluster and itself is zero. The above tables can be compared with those given in Figs 10.4 to 10.6.

such. The k patterns to form clusters may be chosen arbitrarily: they are usually the first k patterns. In this step, the centroid of a given cluster is the coordinates of the pattern in the cluster in the A_1 - A_2 -...- A_M coordinate space.

- For $i = k + 1, k + 2, \dots, r$ do
Put the i th pattern in the cluster whose centroid is the nearest to the pattern. Compute the cluster's new centroid.
- For $i = 1, 2, \dots, r$ do steps 3.1 to 3.3.

- Let C' be the cluster which has the i th pattern. Calculate the distance between the i th pattern and the centroids of each of the k clusters. If the pattern is closest to the centroid of C' , then the pattern need not change its cluster, hence go to step 3.3.
- Let C'' be the cluster whose centroid is the closest to the i th pattern. Move the pattern from cluster C' to cluster C'' . Compute the new centroids of clusters C' and C'' . Go to step 3.
- Continue.

- No patterns changed clusters in the last iteration, hence return from the procedure with the k clusters.

As should be clear from above, the k -means procedure begins by putting each pattern in one of the k clusters (steps 1 and 2). Then, in step 3, it ensures that every pattern is in a cluster whose centroid is closest to it: if moving a pattern from one cluster to another is required, the procedure does so. This makes the k -means procedure preferable to the rest of the clustering procedures. The procedure is said to be *partitional* because it partitions the set of patterns into clusters.

As an example, let us create two clusters of the five patterns X1 to X5 mentioned in step 1 of the k -means procedure, we begin by considering each of the first two patterns to form their own clusters. From Fig. 10.1, we see that the coordinates of the centroids of the two clusters, and the coordinates of the three remaining patterns are

{X1}	{X2}	X3	X4	X5
(1, 1)	(2, 3)	(3, 1)	(4, 4)	(5, 2)

To create two clusters of the five patterns X1 to X5 given in Fig. 10.1 by the **centroid-linkage** procedure using Euclidean distance, we begin by creating 5 clusters {X1}, {X2}, {X3}, {X4}, and {X5}. Considering the off-diagonal values given in Fig. 10.3, we see that the minimum distance between any two clusters is 2: it is between {X1} and {X3}, which we merge into one cluster. The coordinates of the centroid of the resulting cluster {X1, X3} are $[(1+3)/2=2, (1+1)/2=1]$. We then obtain the distances between the four clusters {X1, X3}, {X2}, {X4}, and {X5}, as given in the following symmetric table:

Clusters	{X1, X3}	{X2}	{X4}	{X5}
{X1, X3}	0	2	3.6	3.2
{X2}	2	0	2.2	3.2
{X4}	3.6	2.2	0	2.2
{X5}	3.2	3.2	2.2	0

Since we are using the centroid-linkage procedure, the distance between {X1, X3} and, say, {X4} is $\sqrt{(4-2)^2 + (4-1)^2} = 3.6$, which is what is shown in the table above. The other values in the table are obtained similarly. For simplicity, the values have been rounded off to one place after the decimal. The smallest off-diagonal value in the table is 2; it is between {X1, X3} and {X2}, which we merge into one cluster. The coordinates of the centroid of the resulting cluster {X1, X2, X3} are $[(1+2+3)/3=2, (1+3+1)/3=1.67]$. We now have the three clusters {X1, X2, X3}, {X4}, and {X5}, the distances between which are in the following symmetric table:

Clusters	{X1, X2, X3}	{X4}	{X5}
{X1, X2, X3}	0	3.1	3
{X4}	3.1	0	2.2
{X5}	3	2.2	0

With our using the centroid-linkage procedure, the distance between {X1, X2, X3} and, say, {X4} is $\sqrt{(4-2)^2 + (4-1.67)^2} = 3.1$, as shown in the above table. The smallest off-diagonal value in the table is 2.2. We merge {X5} with {X4} to finally obtain these two clusters: {X1, X2, X3} and {X4, X5}.

FIGURE 10.7 An example of clustering the patterns X1 to X5 given in Fig. 10.1 by the centroid-linkage procedure using Euclidean distance. In the tables, the diagonal elements are zero because the distance between a cluster and itself is zero. The above tables can be compared with those given in Figs 10.4 to 10.6.

such. The k patterns to form clusters may be chosen arbitrarily: they are usually the first k patterns. In this step, the centroid of a given cluster is the coordinates of the pattern in the cluster in the A_1 - A_2 - $\dots$ - A_M coordinate space.

2. For $i = k+1, k+2, \dots, r$ do
Put the i th pattern in the cluster whose centroid is the nearest to the pattern. Compute the cluster's new centroid.
3. For $i = 1, 2, \dots, r$ do steps 3.1 to 3.3.
 - 3.1. Let C' be the cluster which has the i th pattern. Calculate the distance between the i th pattern and the centroids of each of the k clusters. If the pattern is closest to the centroid of C' , then the pattern need not change its cluster, hence go to step 3.3.
 - 3.2. Let C'' be the cluster whose centroid is the closest to the i th pattern. Move the pattern from cluster C' to cluster C'' . Compute the new centroids of clusters C' and C'' . Go to step 3.
 - 3.3. Continue.
4. No patterns changed clusters in the last iteration; hence return from the procedure with the k clusters.

As should be clear from above, the k -means procedure begins by putting each pattern in one of the k clusters (steps 1 and 2). Then, in step 3, it ensures that every pattern is in a cluster whose centroid is closest to it: if moving a pattern from one cluster to another is required, the procedure does so. This makes the k -means procedure preferable to the rest of the clustering procedures. The procedure is said to be *partitional* because it partitions the set of patterns into clusters.

As an example, let us create two clusters of the five patterns X1 to X5 given in Fig. 10.1 by the k -means procedure using Euclidean distance. As mentioned in step 1 of the k -means procedure, we begin by considering each of the first two patterns to form their own clusters. From Fig. 10.1, we see that the coordinates of the centroids of the two clusters, and the coordinates of the three remaining patterns are

{X1}	{X2}	X3	X4	X5
(1, 1)	(2, 3)	(3, 1)	(4, 4)	(5, 2)

We are now in step 2 of the k -means procedure. The distance between

pattern X_3 and the centroid of cluster $\{X_1\}$ is

$$\sqrt{(3-1)^2 + (1-1)^2} = 2$$

The distance between pattern X_3 and the centroid of cluster $\{X_2\}$ is

$$\sqrt{(3-2)^2 + (1-3)^2} = 2.2$$

X_3 is closer to the centroid of $\{X_1\}$ than to $\{X_2\}$; hence we put X_3 into $\{X_1\}$.

The coordinates of the centroid of the resulting cluster $\{X_1, X_3\}$ become $[(1+3)/2 = 2, (1+1)/2 = 1]$. Then the coordinates of the centroids of the

two clusters, and the coordinates of the two remaining patterns are

$$\begin{array}{cc} \{X_1, X_3\} & \{X_2\} & X_4 & X_5 \\ (2, 1) & (2, 3) & (4, 4) & (5, 2) \end{array}$$

The distance between pattern X_4 and the centroid of cluster $\{X_1, X_3\}$ is

$$\sqrt{(4-2)^2 + (4-1)^2} = 3.6$$

The distance between pattern X_4 and the centroid of cluster $\{X_2\}$ is

$$\sqrt{(4-2)^2 + (4-3)^2} = 2.2$$

X_4 is closer to the centroid of $\{X_2\}$ than to $\{X_1, X_3\}$, hence we put X_4

into $\{X_2\}$. The coordinates of the centroid of the resulting cluster $\{X_2, X_4\}$ become $[(2+4)/2 = 3, (3+4)/2 = 3.5]$. The coordinates of the centroids of

the two clusters, and the coordinates of the remaining pattern are then

$$\begin{array}{cc} \{X_1, X_3\} & \{X_2, X_4\} & X_5 \\ (2, 1) & (3, 3.5) & (5, 2) \end{array}$$

The distance between pattern X_5 and the centroid of cluster $\{X_1, X_3\}$ is

$$\sqrt{(5-2)^2 + (2-1)^2} = 3.2$$

The distance between pattern X_5 and the centroid of cluster $\{X_2, X_4\}$ is

$$\sqrt{(5-3)^2 + (2-3.5)^2} = 2.5$$

X_5 is closer to the centroid of $\{X_2, X_4\}$ than to $\{X_1, X_3\}$, hence we put X_5 into $\{X_2, X_4\}$. The coordinates of the centroid of the resulting cluster $\{X_2, X_4, X_5\}$ become $[(2+4+5)/3 = 3.67, (3+4+2)/3 = 3]$. The coordinates of the two clusters are accordingly

$$\begin{array}{cc} \{X_1, X_3\} & \{X_2, X_4, X_5\} \\ (2, 1) & (3.67, 3) \end{array}$$

Each of the five patterns is now in a cluster. We are now in step 3 of the k -means procedure. We next need to iterate through the patterns and move each pattern from the cluster it is in to the cluster whose centroid is nearest to the pattern. Accordingly, we notice that the distance between X_1 and the centroid of cluster $\{X_1, X_3\}$ is

$$\sqrt{(1-2)^2 + (1-1)^2} = 1$$

The distance between X_1 and the centroid of cluster $\{X_2, X_4, X_5\}$ is

$$\sqrt{(1-3.67)^2 + (1-3)^2} = 3.3$$

X_1 is closer to the centroid of $\{X_1, X_3\}$ than to $\{X_2, X_4, X_5\}$; hence X_1 stays in the cluster it is in. We next notice that the distance between X_2 and the centroid of cluster $\{X_1, X_3\}$ is

$$\sqrt{(2-2)^2 + (3-1)^2} = 2$$

The distance between X_2 and the centroid of cluster $\{X_2, X_4, X_5\}$ is

$$\sqrt{(2-3.67)^2 + (3-3)^2} = 1.67$$

X_2 is closer to the centroid of $\{X_2, X_4, X_5\}$, than to $\{X_1, X_3\}$; hence X_2 stays in the cluster it is in. Proceeding along, we notice that the distance between X_3 and the centroid of cluster $\{X_1, X_3\}$ is

$$\sqrt{(3-2)^2 + (1-1)^2} = 1$$

The distance between X_3 and the centroid of cluster $\{X_2, X_4, X_5\}$ is

$$\sqrt{(3-3.67)^2 + (1-3)^2} = 2.1$$

We are now in step 2 of the k -means procedure. The distance between pattern X3 and the centroid of cluster {X1} is

$$\sqrt{(3-1)^2 + (1-1)^2} = 2$$

The distance between pattern X3 and the centroid of cluster {X2} is

$$\sqrt{(3-2)^2 + (1-3)^2} = 2.2$$

X3 is closer to the centroid of {X1} than to {X2}; hence we put X3 into {X1}. The coordinates of the centroid of the resulting cluster {X1, X3} become $[(1+3)/2 = 2, (1+1)/2 = 1]$. Then the coordinates of the centroids of the two clusters, and the coordinates of the two remaining patterns are

{X1, X3}	{X2}	X4	X5
(2, 1)	(2, 3)	(4, 4)	(5, 2)

The distance between pattern X4 and the centroid of cluster {X1, X3} is

$$\sqrt{(4-2)^2 + (4-1)^2} = 3.6$$

The distance between pattern X4 and the centroid of cluster {X2} is

$$\sqrt{(4-2)^2 + (4-3)^2} = 2.2$$

X4 is closer to the centroid of {X2} than to {X1, X3}, hence we put X4 into {X2}. The coordinates of the centroid of the resulting cluster {X2, X4} become $[(2+4)/2 = 3, (3+4)/2 = 3.5]$. The coordinates of the centroids of the two clusters, and the coordinates of the remaining pattern are then

{X1, X3}	{X2, X4}	X5
(2, 1)	(3, 3.5)	(5, 2)

The distance between pattern X5 and the centroid of cluster {X1, X3} is

$$\sqrt{(5-2)^2 + (2-1)^2} = 3.2$$

The distance between pattern X5 and the centroid of cluster {X2, X4} is

$$\sqrt{(5-3)^2 + (2-3.5)^2} = 2.5$$

X5 is closer to the centroid of {X2, X4} than to {X1, X3}, hence we put X5 into {X2, X4}. The coordinates of the centroid of the resulting cluster {X2, X4, X5} become $[(2+4+5)/3 = 3.67, (3+4+2)/3 = 3]$. The coordinates of the centroids of the two clusters are accordingly

{X1, X3}	{X2, X4, X5}
(2, 1)	(3.67, 3)

Each of the five patterns is now in a cluster. We are now in step 3 of the k -means procedure. We next need to iterate through the patterns and move each pattern from the cluster it is in to the cluster whose centroid is nearest to the pattern. Accordingly, we notice that the distance between X1 and the centroid of cluster {X1, X3} is

$$\sqrt{(1-2)^2 + (1-1)^2} = 1$$

The distance between X1 and the centroid of cluster {X2, X4, X5} is

$$\sqrt{(1-3.67)^2 + (1-3)^2} = 3.3$$

X1 is closer to the centroid of {X1, X3} than to {X2, X4, X5}; hence X1 stays in the cluster it is in. We next notice that the distance between X2 and the centroid of cluster {X1, X3} is

$$\sqrt{(2-2)^2 + (3-1)^2} = 2$$

The distance between X2 and the centroid of cluster {X2, X4, X5} is

$$\sqrt{(2-3.67)^2 + (3-3)^2} = 1.67$$

X2 is closer to the centroid of {X2, X4, X5}, than to {X1, X3}; hence X2 stays in the cluster it is in. Proceeding along, we notice that the distance between X3 and the centroid of cluster {X1, X3} is

$$\sqrt{(3-2)^2 + (1-1)^2} = 1$$

The distance between X3 and the centroid of cluster {X2, X4, X5} is

$$\sqrt{(3-3.67)^2 + (1-3)^2} = 2.1$$

X3 is closer to the centroid of {X1, X3} than to {X2, X4, X5}; hence X3 stays in the cluster it is in. In the same vein, we notice that the distance between X4 and the centroid of cluster {X1, X3} is

$$\sqrt{(4-2)^2 + (4-1)^2} = 3.6$$

The distance between X4 and the centroid of cluster {X2, X4, X5} is

$$\sqrt{(4-3.67)^2 + (4-3)^2} = 1.1$$

X4 is closer to the centroid of {X2, X4, X5} than to {X1, X3}; hence X4 stays in the cluster it is in. Finally, we notice that the distance between X5 and the centroid of cluster {X1, X3} is

$$\sqrt{(5-2)^2 + (2-1)^2} = 3.2$$

The distance between X5 and the centroid of cluster {X2, X4, X5} is

$$\sqrt{(5-3.67)^2 + (2-3)^2} = 1.7$$

Procedure	The two clusters obtained
Single-linkage	{X1, X2, X3, X4} and {X5}
Complete-linkage	
Average-linkage	
Centroid-linkage	{X1, X2, X3} and {X4, X5}
k-means	{X1, X3} and {X2, X4, X5}

FIGURE 10.8

Recapitulating the results of clustering the five patterns given in Fig. 10.1 into two clusters by different procedures, using Euclidean distance. Not all procedures give the same final clusters. The results of the different linkage procedures have been obtained from Figs 10.4 to 10.7. The results of the k-means procedure are from the example solved in Section 10.3. The complete-linkage, average-linkage, and centroid-linkage procedures have coincidentally produced identical clusters¹.

¹The author is not aware of any theory that says the three procedures will always produce identical clusters from a given set of patterns.

X5 is closer to the centroid of {X2, X4, X5} than to {X1, X3}; hence X5 stays in the cluster it is in. We have completed an iteration, and no pattern changed its cluster; hence we terminate with the clusters as they exist: {X1, X3} and {X2, X4, X5}. Had any pattern changed its cluster, we would need to recalculate the centroids of the newly formed clusters and iterate again. For the sake of brevity, this example terminates after one iteration. In practice, many such iterations may be required.

The clusters resulting from the patterns given in Fig. 10.1, using the five clustering procedures discussed earlier (single-linkage, complete-linkage, average-linkage, centroid-linkage, and k-means), are recapitulated in Fig. 10.8 for the convenience of the readers.

In calculating Euclidean distances while clustering, finding the square roots can be omitted to save computation. It will not affect the results, square root being monotonic. Nonetheless, in the preceding examples, the square roots have been calculated to avoid any confusion in understanding the procedures.

10.4 Non-numeric Attributes

Clustering is mostly carried out on patterns with numeric attributes. With non-numeric attributes, it is difficult to decide into which cluster a pattern should be put, because it is difficult to measure the similarity between patterns. Nonetheless, clustering can be carried out by using a measure such as Hamming distance. As mentioned in Section 6.3, the Hamming distance between two patterns is the number of attributes in which the two patterns have different values. Figure 10.9 shows the clustering of the professor-student patterns given in Fig. 1.1 by the average-linkage procedure using Hamming distance. The figure shows that the two clusters obtained are {T1, T2, T3, T4, T6, T7} and {T5, T8}. It is debatable whether the two clusters are meaningful. They are not the same as the classes of the patterns.

X3 is closer to the centroid of {X1, X3} than to {X2, X4, X5}; hence X3 stays in the cluster it is in. In the same vein, we notice that the distance between X4 and the centroid of cluster {X1, X3} is

$$\sqrt{(4-2)^2 + (4-1)^2} = 3.6$$

The distance between X4 and the centroid of cluster {X2, X4, X5} is

$$\sqrt{(4-3.67)^2 + (4-3)^2} = 1.1$$

X4 is closer to the centroid of {X2, X4, X5} than to {X1, X3}; hence X4 stays in the cluster it is in. Finally, we notice that the distance between X5 and the centroid of cluster {X1, X3} is

$$\sqrt{(5-2)^2 + (2-1)^2} = 3.2$$

The distance between X5 and the centroid of cluster {X2, X4, X5} is

$$\sqrt{(5-3.67)^2 + (2-3)^2} = 1.7$$

Procedure	The two clusters obtained
Single-linkage	{X1, X2, X3, X4} and {X5}
Complete-linkage	{X1, X2, X3} and {X4, X5}
Average-linkage	
Centroid-linkage	
k-means	{X1, X3} and {X2, X4, X5}

FIGURE 10.8 Recapitulating the results of clustering the five patterns given in Fig. 10.1 into two clusters by different procedures, using Euclidean distance. Not all procedures give the same final clusters. The results of the different linkage procedures have been obtained from Figs 10.4 to 10.7. The results of the *k*-means procedure are from the example solved in Section 10.3. The complete-linkage, average-linkage, and centroid-linkage procedures have coincidentally produced identical clusters¹.

¹The author is not aware of any theory that says the three procedures will always produce identical clusters from a given set of patterns.

X5 is closer to the centroid of {X2, X4, X5} than to {X1, X3}; hence X5 stays in the cluster it is in.

We have completed an iteration, and no pattern changed its cluster; hence we terminate with the clusters as they exist: {X1, X3} and {X2, X4, X5}. Had any pattern changed its cluster, we would need to recalculate the centroids of the newly formed clusters and iterate again. For the sake of brevity, this example terminates after one iteration. In practice, many such iterations may be required.

The clusters resulting from the patterns given in Fig. 10.1, using the five clustering procedures discussed earlier (single-linkage, complete-linkage, average-linkage, centroid-linkage, and *k*-means), are recapitulated in Fig. 10.8 for the convenience of the readers.

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